Accelerating Cohort Identification from EHRs with Biomedical Knowledge Driven AI Agents

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Introduction

Identifying patient cohorts is a crucial first step in numerous medical research and healthcare analytics endeavors. This process enables researchers to monitor disease progression, identify risk factors, and evaluate the long-term efficacy and safety of medical interventions [1]. Accordingly, there is a continual demand for accurate and efficient methods to identify cohorts from the vast populations represented in Electronic Health Record (EHR) systems. Yet, this task remains challenging for both biomedical research and the generation of real-world evidence due to the heterogeneous structure of EHR data and the inconsistent representation of diseases, drugs, and phenotypes. The pursuit of efficient and accurate cohort identification has therefore driven decades of innovation in clinical informatics. The technological landscape has evolved considerably, progressing from early rule-based text processing to sophisticated, standardized data platforms and, most recently, to the paradigm-shifting capabilities of large language models (LLMs). Meanwhile, harmonization of biomedical knowledge through ontologies such as the Disease Ontology and RxNorm addresses the complementary challenge of ensuring consistent semantic alignment across structured and unstructured data sources. Taken together, LLMs offer the power to interpret complex language in eligibility criteria, while ontologies provide the standardized backbone needed for interoperability and reproducibility. In this work, we present an agent-based AI framework that integrates LLMs with biomedical knowledge to accelerate cohort identification while ensuring semantic interoperability.

Method and Results

As shown in Figure 1, to enable accurate and efficient patient cohort identification from heterogeneous EHR data, our system integrates biomedical knowledge resources with two key LLM-based agents that jointly address structured and unstructured EHR data types:

Text-to-SQL Agent: This agent focuses on structured data queries. It transforms natural language eligibility criteria into executable SQL over an OMOP CDM–compliant database. To ensure semantic consistency, disease and drug terms appearing in criteria are normalized using standardized biomedical vocabularies such as the Disease Ontology, RxNorm, and SNOMED CT. Equipped with tools including a SQL schema navigator, SQL runner, and ontology-enhanced phenotype search modules, the agent enables researchers to filter cohorts without requiring manual SQL coding, while ensuring interoperability across datasets.

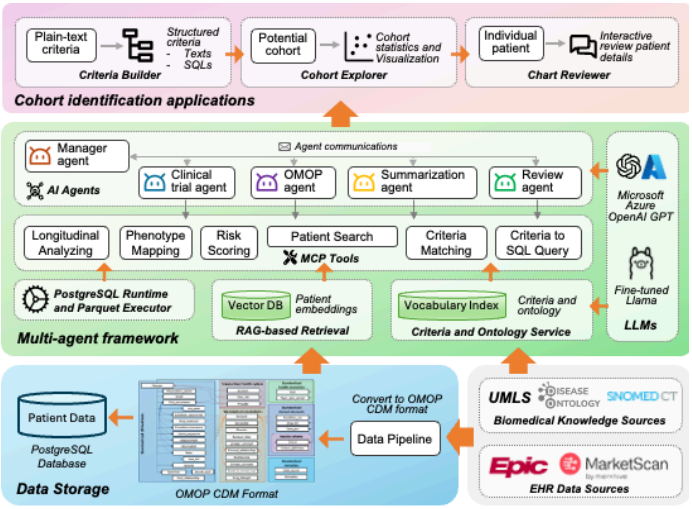

Figure 1: The overall architecture of the proposed framework

Criteria Agent: This agent addresses complex eligibility conditions that are difficult to define purely in SQL. It leverages a criteria library service that indexes inclusion and exclusion conditions enriched with ontology mappings to ICD and SNOMED CT concepts. To match patient data with free-text criteria, the agent employs criteria search, note embedding search, and clinical fact retrieval modules, all augmented by ontology-based synonym expansion and hierarchical reasoning. This design allows the agent to extract subtle information from unstructured sources such as clinical notes and pathology reports, and to provide both a decision and supporting evidence mapped back to standardized biomedical knowledge. We will demonstrate the feasibility and efficiency of these agents and open-source the agent design at https://github.com/BIDS-Xu-Lab/echo-criteria-agents.

Discussion and Future Work

While still under development, the prototype highlights how agent-driven services can lower technical barriers and improve efficiency in trial recruitment. Future work will focus on expanding agent scalability, conducting formal evaluations, and integrating clinician feedback to ensure real-world applicability.

References

1. Shivade, C. et al (2014). A review of approaches to identifying patient phenotype cohorts using electronic health records. *Journal of the American Medical Informatics Association : JAMIA*, *21*(2), 221–230.